NAME: ______________________ **PERIOD:** ________ **DATE:** ____________

Fourteen Caption Users

AP Statistics · Unit 3 · Topic 3.2 · B: 2026-12-01 · E: 2026-12-02

[STAR] TODAY'S PROBLEM

Suppose a streaming platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\widehat{p} = 0.14$.

Jules: "Because $n = 100 > 30$, $\widehat{p}$ is approximately normal. I get $P(\widehat{p} \geq 0.14) = 0.0135$, which proves caption use increased."

Nora: "Because $np = 8 < 10$, a normal model is not justified. Therefore $\widehat{p}$ is biased and this sample tells us nothing."

A binomial calculator gives $P(X \geq 14) = 0.0282$ for $X \sim \text{Bin}(100, 0.08)$. This closely approximates sampling from 5,000 accounts.

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE — before we discuss (5 min)

Does finding 14 caption users instead of the expected 8 prove that use increased? Give one reason.

[BOOK] CHECK INDEPENDENCE, THEN SHAPE (keep on the page)

For an SRS, $\mu_{\widehat{p}} = p$ and $\sigma_{\widehat{p}} = \sqrt{p(1-p)/n}$ when $n < 10\%$ of the population makes without-replacement dependence negligible. Approximate normality is a separate question: require both $np \geq 10$ and $n(1-p) \geq 10$; a generic $n > 30$ rule is not enough for proportions. If Large Counts fails, use an exact binomial calculation when independence is reasonable or simulate the sampling process.

Work (a)–(c) from the rules box:

DOK 1 **(a)** Identify p and verify the sample proportion $\widehat{p} = 14/100$.

DOK 2 **(b)** Find the mean and standard deviation of the sampling distribution of $\widehat{p}$. Check the 10 percent condition and both Large Counts values.

DOK 3 **(c)** In 3–4 sentences, explain what each student gets right and wrong. Use the condition checks and the supplied probability 0.0282. State what the sample suggests about caption use and why a failed normal-shape check does not make the estimate biased.

Frame: The condition ____________ passes, but ____________ fails.

Frame: The probability 0.0282 suggests __________________; failure of normal shape does not mean __________________.

EXIT — turn this in

One thing the rules box changed about my first take: ________________________